Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal

DLA - Microcirculation

St. George's University
Grenada, West Indies

Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0128	Differentiate the following terms: osmotic pressure, oncotic pressure, and hydrostatic pressure, as they pertain to movement across the endothelium of the capillaries.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0129	Based on the Starling hypothesis, explain how permeability, hydrostatic pressure and oncotic pressure influence transcapillary exchange of fluid
SOM.MK.II.BPM1.3.CPR.2.PHYS.0130	Describe how changes in capillary surface area affect the capacity for fluid exchange.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0131	Explain the Starling equation and how each component influences fluid movement across the capillary wall
SOM.MK.II.BPM1.3.CPR.2.PHYS.0132	Predict how altering pressure or resistance in pre- and post-capillary regions alters capillary pressure and the consequence of this change on transmural fluid movement.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0133	Discuss the lymphatics and how the structural characteristics allow the reabsortion of large compounds including proteins
SOM.MK.II.BPM1.3.CPR.2.PHYS.0134	Contrast the structure of lymphatic capillaries and systemic capillaries, including the significance of the smooth muscle in the walls of the lymphatic vessels.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0135	Identify critical functions of the lymphatic system in fat absorption, interstitial fluid reabsorption, and clearing large proteins from the interstitial spaces.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0136	Discuss the relationship between the interstitial pressure and the lymph flow, and why an increase in interstitial pressure does not normally cause edema.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0137	Explain how edema develops in response to: a) venous obstruction, b) lymphatic obstruction, c) increased capillary permeability, d) heart failure, e) tissue injury or allergic reaction, and f) malnutrition.

Recommended reading

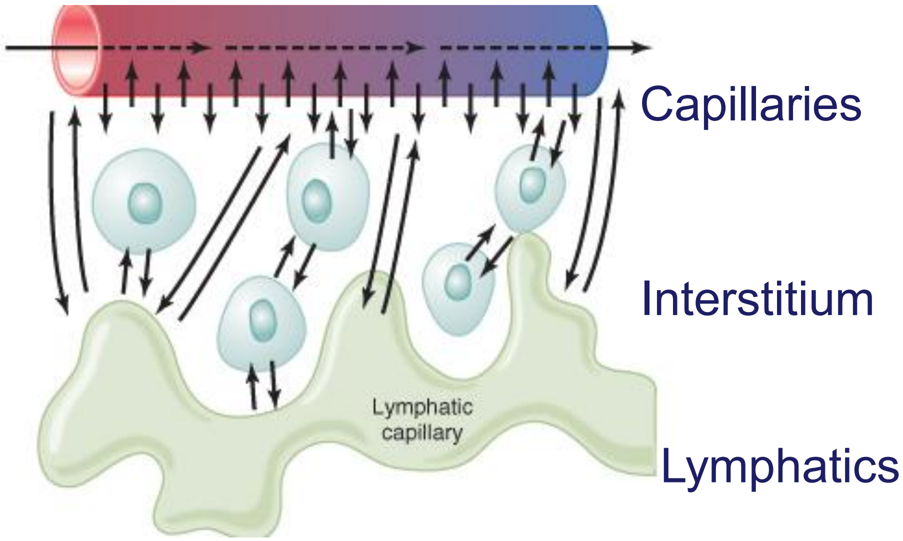
1. Medical Physiology: Principles for Clinical Medicine, 6e.

Rodney A. Rhoades, David R. Bell. Chapter 15: Regulation of Organ Blood Flow and Capillary Transport – The microcirculation and capillary dynamics

2. Cardiovascular Physiology Concepts, 3e. Richard E. Klabunde. Chapter 8: Exchange function of Microcirculation

Microcirculation
- arteries, venules, capillaries, interstitium and lymphatics

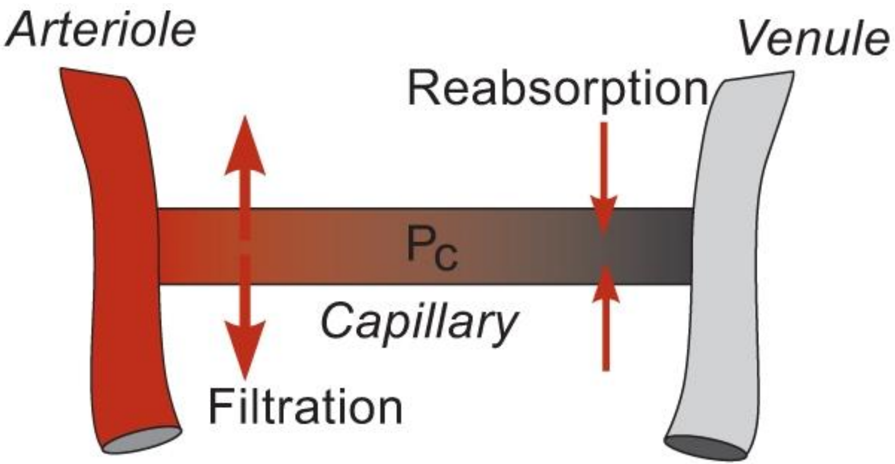
Microcirculation is the site of exchange of nutrients, water, gases, and small molecules between the plasma and the tissues. The exchange is between the interstitial fluid and capillaries.
Exchange occurs via three processes: (1) diffusion, (2) filtration, and (3) vesicular transport.



1. **Hydrostatic pressure:**
   Capillary pressures drive fluid out of capillaries into interstitial space
2. **Oncotic/colloid pressure:**
   Capillary oncotic pressures keep fluid in capillaries and drive reabsorption.
3. **Capillary walls:** permeable
4. **Lymphatic drainage:** Remove excess interstitial fluids (related to Starling hypothesis)

Starlings' hypothesis and capacity for fluid exchange

Starling's hypothesis: "under normal conditions, the amount of fluid filtering outward from the arterial ends of the capillaries equals almost exactly the fluid returned to the circulation by absorption".



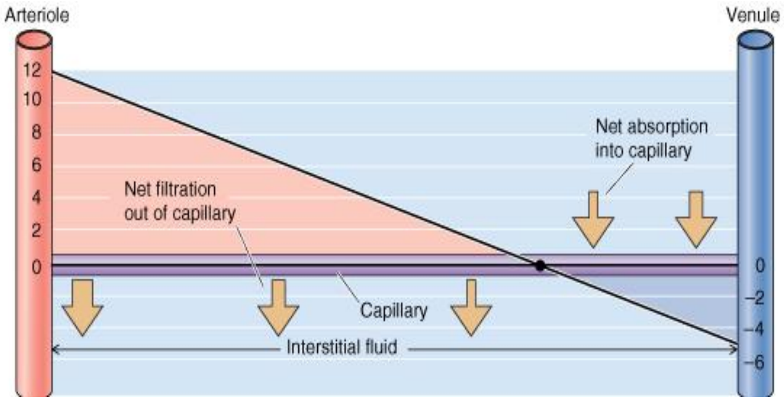
Modified Klabunde 2nd ed Cardiovascular Physiol Concepts

The microcirculation regulates blood flow and blood flow distribution to individual organs.

Blood supply to capillaries is controlled by intermittent contraction and relaxation of muscles in metarterioles and precapillary sphincters (Vasomotion).

By controlling the number of capillaries open to flow at any given time (in most organs), it effects changes in diffusion distances between an organ's blood supply and tissues, as well as the capillary surface area available for exchange of materials between the plasma and tissues

Starling-Landis Equation
- Calculating Net Filtration Pressure (NFP) and Flux

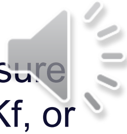


- $J_v = K_f[(P_c - P_i) - (\pi_c - \pi_i)]$

Modified Boron and Boulapaep

- fluid movement is determined by the difference in the sum of hydrostatic and oncotic pressures

- **variables**
- Kf -"hydraulic conductance" or water permeability of the capillary wall. It determines the magnitude of fluid movement
- P_c capillary hydrostatic pressure favors filtration out of the capillary
- Pi interstitial hydrostatic pressure opposes filtration
- π_c capillary oncotic pressure opposes filtration
- osmotic pressure of capillary blood due to plasma proteins
- π_i interstitial oncotic pressure favors filtration



- the magnitude of fluid movement for a given pressure difference determined by hydraulic conductance (Kf, or water permeability) of the capillary wall

Net Filtration Pressure (NFP) and Flux:
Starling-Landis Equation

Net Filtration Pressure (NFP)
= ΔHydrostatic pressure –Δoncotic pressure

$$NFP = (P_c - P_{IF}) - \sigma(\pi_c - \pi_{IF})$$

$$J_V = K \; [(P_c - P_{IF}) - \sigma(\pi_c - \pi_{IF})]$$

Capillary pressures placed first in the equation, as they are normally higher!

Jv= flux or net volume of fluid (μm^3/min)

NFP = Net Filtration Pressure (mm HG)
K =hydraulic conductivity and Area
σ = reflection coeff for plasma proteins- protein permeability. (normally constant- changes with hypoxia, injury, inflammation)

1. Starling Equation: positive value = filtration

$$NFP = (P_C - P_{IF}) - \sigma(\pi_C - \pi_{IF})$$

At arteriolar end of capillary

P_C = 35mmHg P_{IF} = -2mmHg

π_C = 25mmHg π_{IF} = 0mmHg

$$NFP = (35-(-2)) - (25-0)$$

$$NFP = 12\ mmHg$$

Positive sign! Filtration has taken place

Examples:

high filtration: glomerulus, retinal capillaries $\uparrow P_C$

lower filtration: lungs $\downarrow P_C$ to reduce edema

2. Starling Equation: negative value = absorption

$$NFP = (P_C - P_{IF}) - \sigma(\pi_C - \pi_{IF})]$$

At venous end of capillary

P_C = 15mmHg P_{IF} = -2mmHg

π_C = 25mmHg π_{IF} = 3mmHg

NFP= (15-(-2)) – (25-3)

(17-22)

NFP= -5 mmHg

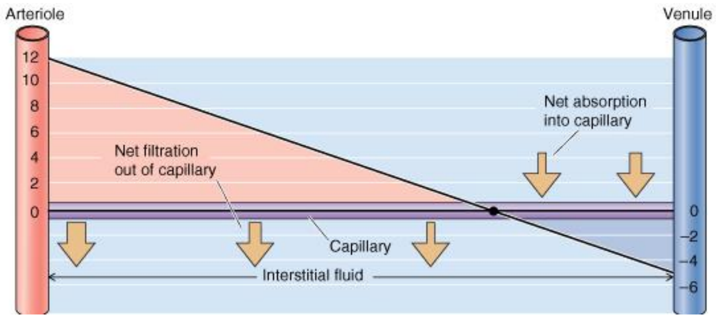
Negative sign! Absorption is taking place

Note: the fall in capillary hydrostatic pressure and rise in IF oncotic pressure !

Net Filtration Pressures:

Altering pressure in pre- and post- capillary region

Filtration = Positive value



Absorption: Negative value

Boron and Boulpaep

Filtration arteriolar end of capillaries > reabsorption at venular end

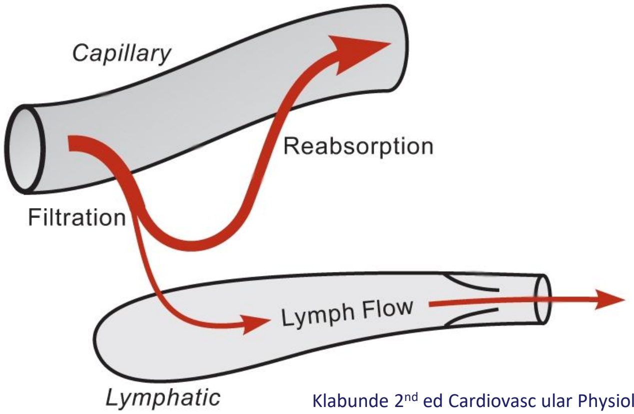
- If the changes in pre- and post-capillary pressures or resistances result in an increased net hydrostatic pressure within the capillaries, more fluid will be pushed out of the capillaries into the interstitial space, leading to edema (accumulation of fluid in tissues).

- Conversely, if the changes lead to an increased oncotic pressure within the capillaries (due to increased protein concentration), more fluid will be drawn into the capillaries from the interstitial space, reducing edema.

alterations in pressure or resistance in pre- and post-capillary regions can modulate capillary pressure, influencing the direction and magnitude of transmural fluid movement across the capillary wall, which ultimately impacts tissue fluid balance and the development of edema

Lymphatics

- Majority of fluid filtered at the arterial end is reabsorbed at the venous end.
- Filtration arteriolar end of capillaries > reabsorption at venous end: ~2-4 L/day.
- Starling hypothesis a bit off with "filtration almost exactly = reabsorption"
- However fluid does not normally accumulate in interstitium



Klabunde 2nd ed Cardiovasc ular Physiol Concepts

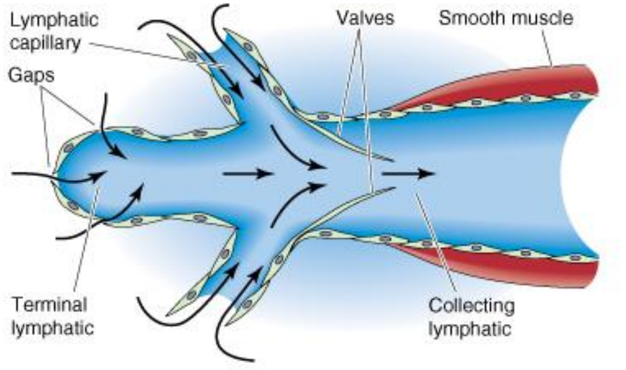
- Fluid not reabsorbed in capillaries is drained by the lymphatic system.
  -(~5-10% of filtrate transported out of tissues by lymphatics)

- Proteins and bacteria removed.

The lymphatic system helps maintain fluid balance by collecting excess interstitial fluid (lymph) from tissues and returning it to the bloodstream.

This prevents the accumulation of excess fluid in tissues, thereby preventing edema (swelling).

Lymphatics: movement of lymph



P_{lymph} low

P_{lymph} higher after valves

Guyton and Hall

Hydrostatic pressures

Lymphatic (P_{lymph}) **<** Interstitium (P_{if})

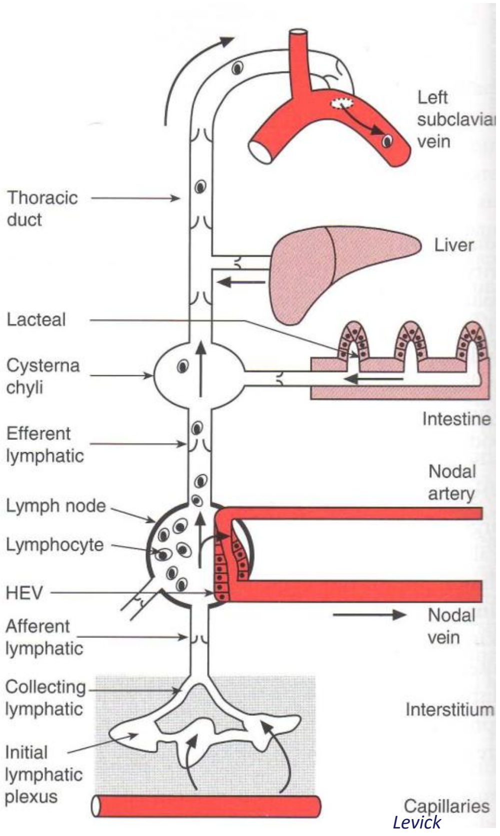
<u>But</u> P_{lymph} can increase due to pumping in lymph vessels (after valves)

- Transient increases in P_{if} drive fluid into lymphatics. Interstitial volume normally kept relatively constant.
- Pumping action of open terminal ends of lymphatics draw fluid into lymphatics (Lymph P >P_{if}). Pumping inc when more fluid in P_{if}

Movement of lymph:

- Valves in lymphatics → one way movement of lymph.
- Contraction/Pumping: Movement of lymphatic fluids. Pressures in larger lymphatics can be 100 mm Hg.
- Compression: by skeletal muscles, respiratory movements and intestinal contractions.

Lymphatic System



Lymphatic capillaries merge into the thoracic duct.

Thoracic duct drains lymph into the left subclavian vein.

Lymphatic drainage reduces [protein] in interstitium- controls interstitium colloid pressures.

Fat absorption will be covered in GI system.

Uptake of lymphatic fluid resists edema formation

EDEMA

a.Most causes of **edema** can be explained in terms of changes in one or more of the variables:

a. The most common causes of edema result from **increased capillary hydrostatic pressures. E.g.** due to heart failure or venous obstruction.
   - pulmonary edema: slight increases in hydrostatic pressures in lung capillaries.
   - peripheral edema: Inc in systemic capillary pressures result in inc filtration in lower extremities and intestine

b. **Generalized edema** may also result from low serum protein concentration due to reduced capillary oncotic pressure (π_c). For example, this can occur in liver failure when insufficient albumin is synthesized, in pregnancy, malnutrition and in nephrotic syndrome when plasma protein is excreted in the urine.

c. Increased capillary permeability: seen in **inflammation or sepsis, ishmeia and burns.** Generalized edema may occur because inflammatory mediators reduce protein reflection coefficient (σ) and proteins leak into the interstitium, thereby increasing interstitial fluid oncotic pressure

d. **Lymphatic blockage** causes edema because the tissue fluid that has formed is trapped in the interstitium when the outflow pathway is blocked. Impaired e.g. parasites, surgical lymph node removal. Local edema before blocked nodes

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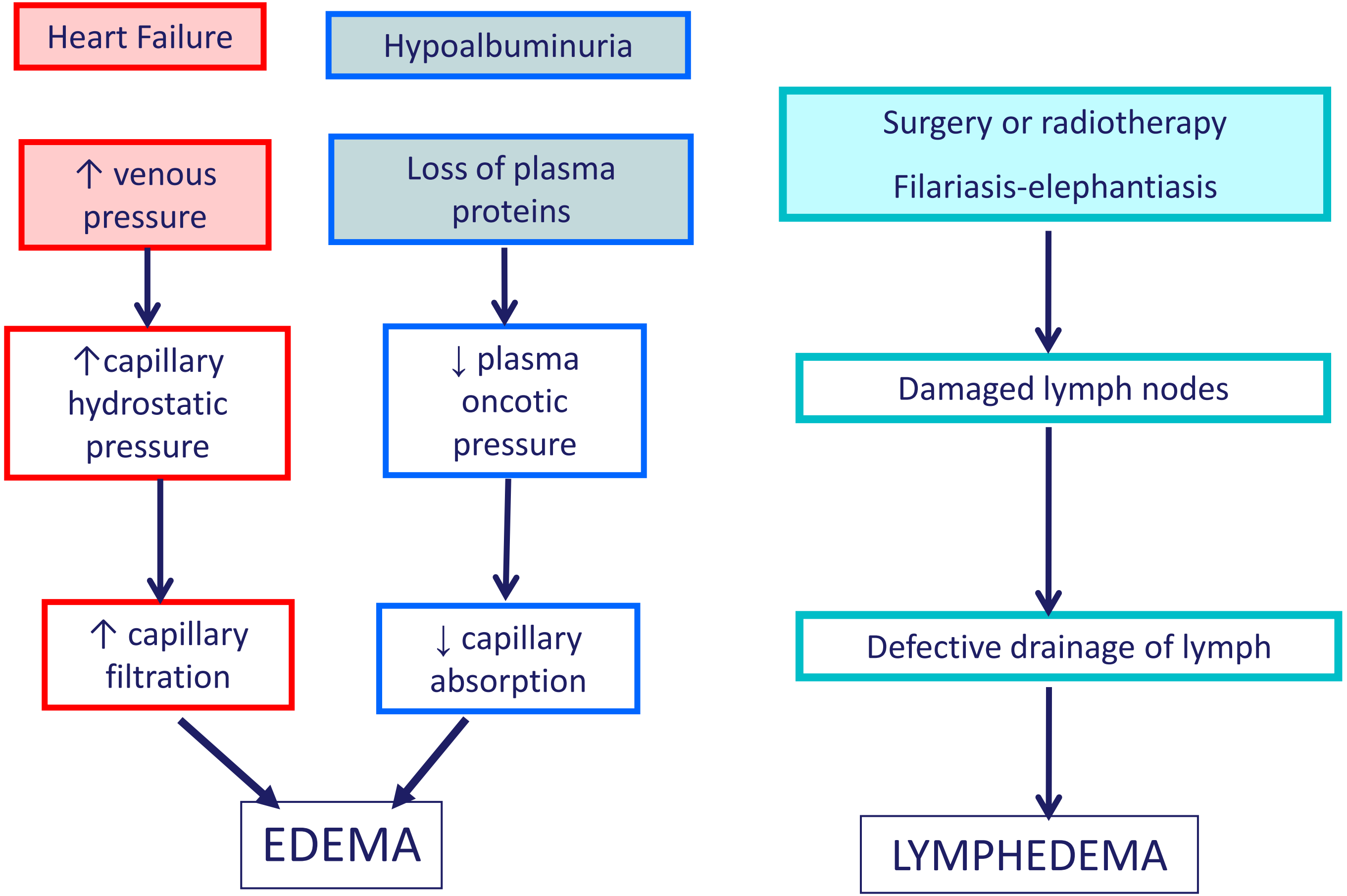
Heart Failure
↑ venous pressure
↑capillary hydrostatic pressure
↑ capillary filtration
Hypoalbuminuria
Loss of plasma proteins
↓ plasma oncotic pressure
↓ capillary absorption
EDEMA
Surgery or radiotherapy
Filariasis-elephantiasis
Damaged lymph nodes
Defective drainage of lymph
LYMPHEDEMA

EDEMA Causes: Organ dysfunction

<u>Right Ventricular Failure</u>

↑ Right Atrial Pressure

↓

↑ CVP

↓

↑ Capillary Pressure (systemic)

↓

↑ Filtration

↓

Tissue Edema

↓

Ankle Swelling (pitting edema)

<u>Left Ventricular Failure</u>

Can result in pulmon hypertension

↑ Left Atrial Pressure

↓

↑ Pulmonary Venous Pressure

↓

↑ Capillary Pressure (pulmonary)

↓

↑ Filtration

↓

Pulmonary Edema

↓

Fluid in the lungs

<u>Hepatic fibrosis</u> → ↑ hepatic venous pressure

↙

Ascites: accumulation of fluid in peritoneal cavity

Practice MCQs

185. A 62-year-old man with congestive heart failure (CHF) develops increasing shortness of breath in the recumbent position. A chest x-ray reveals cardiomegaly, horizontal lines perpendicular to the lateral lung surface indicative of increased opacity in the pulmonary septa, and lung consolidation. Pulmonary edema in CHF is promoted by which of the following?

a. Decreased pulmonary capillary permeability
b. Decreased pulmonary interstitial oncotic pressure
c. Increased pulmonary capillary hydrostatic pressure
d. Increased pulmonary capillary oncotic pressure
e. Increased pulmonary interstitial hydrostatic pressure

202. A 65-year-old smoker develops a squamous cell bronchogenic carcinoma that metastasizes to the tracheobronchial and parasternal lymph nodes. The chest x-ray is consistent with accumulation of fluid in the pulmonary interstitial space. Flow of fluid through the lymphatic vessels will be decreased if there is an increase in which of the following?

a. Capillary oncotic pressure
b. Capillary permeability
c. Capillary pressure
d. Central venous pressure
e. Interstitial protein concentration

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Questions sourced from Pret-Test for Physiology, 14th edition 2014

Any Questions? Come by my office, meet on zoom or email!

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